Anand Nagaraj

Design Verification Engineer · UVM · SVA · cocotb · Coverage Closure

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PROFILE

M.Sc. student in Design and Test of Integrated Circuits at TU Chemnitz with **3.5 years of industry experience** in functional verification of digital IP: UVM environments, SystemVerilog Assertions for protocol and CDC checks, Python/cocotb co-simulation and automated regressions with coverage tracking – on NoC, PCIe and RISC-V designs. Seeking a **working student position (up to 20 h/week) or internship** in digital verification.

PROFESSIONAL EXPERIENCE

RTL Design & Verification Engineer (founding team) 11/2024 – 02/2026
Semiconductor IP start-up Bengaluru, India

- Built **UVM and cocotb environments** to verify multi-port Network-on-Chip switches under heavy synthetic traffic.
- Developed scoreboards, transaction checkers and monitors that detect functional mismatches, packet loss and routing-latency violations.
- Wrote **SVA for clock-domain crossings** on asynchronous-FIFO boundaries, proving deadlock-free arbitration and protocol compliance.
- Created a **Python regression framework**: automated test runs, transcript aggregation and functional-coverage extraction; AI-assisted error-log clustering for faster root-cause analysis.
- Verified custom RISC-V (Ibex) instruction extensions against the Spike ISS.

RTL Design & Verification Engineer 09/2022 – 10/2024
Hardware engineering services company Kochi / Bengaluru, India

- Developed modular **UVM components** (agents, drivers, monitors, scoreboards) for a PCIe Transaction Layer IP: TLP formatting, ordering and buffer management.
- Wrote scoreboard checkers for credit-based flow control, targeting corner cases such as credit starvation and back-to-back bursts.
- Built cocotb co-simulation tests for register read/write and hardware/software interface boundaries.
- Authored verification plans and reports; also designed parts of the RTL (TLP parsing, ECRC, flow control).

KEY PROJECTS

Moola-V (MR5) – RV32IM core verification
cocotb · Spike ISS · SVA · RISCOF

- Self-checking cocotb environment running instruction-by-instruction lock-step against the Spike golden ISS.
- SVA and covergroups on pipeline-stage validity, hazard stalls and exceptions; regression suite for every feature change.
- **100 % compliance on all 49 RISCOF suites** (RV32I, RV32M). Core designed from scratch in SystemVerilog. *github.com/AnandNag003/moola_riscv_core*

Post-quantum TLS retrofit on a RISC-V board – NextSEMI 2026 hackathon
C · OpenSSL 3.5 · ML-KEM · ML-DSA

- Retrofitted a TLS 1.3 client on an embedded RISC-V (RV64GC) Linux board with hybrid key exchange (X25519 + ML-KEM-768) and hybrid authentication (ECDSA P-256 + ML-DSA-44).
- Offset the post-quantum overhead with RISC-V bit-manipulation (Zba/Zbb) builds, multithreaded signature verification and CPU-frequency tuning: handshake **31 % faster than the legacy baseline** (14.5 vs 21.0 ms, 1,000 runs, own test server).

AMBA AHB-to-APB bridge – UVM verification suite
SystemVerilog · UVM · SVA

- UVM environment with AHB master and APB slave agents, scoreboard and constrained-random sequences.
- SVA protocol checks and functional coverage for bursts, wait states, error responses and timeouts – **100 % functional and code coverage**.
- Bridge RTL designed by me: FSM control, address decoding, HREADYOUT wait-state propagation.

Dual-port synchronous SRAM macro – RTL to GDSII
Verilog · SkyWater 130 nm · OpenLane 2 · OpenROAD

- Parameterised 16 × 8 dual-port SRAM with byte write-masking and concurrent read/write collision arbitration (Tiny Tapeout).
- Full open-source flow RTL → GDSII: **DRC/LVS clean, STA sign-off**; 6T bit-cell sizing and SNM analysis in SPICE. *github.com/AnandNag003/tt_moola_dp_sram*

Mixed-signal frequency measurement system (TU Chemnitz)
VHDL · VHDL-AMS

- Dual-clock measurement unit with cooperating FSMs, double-flop synchronisers and request/acknowledge CDC handshakes.
- VHDL-AMS model of a MEMS sensor front-end (RC filter, VCO) with threshold logic and 115200-baud UART for end-to-end verification.

EDUCATION

M.Sc. Design and Test of Integrated Circuits 04/2026 – present
Technische Universität Chemnitz Chemnitz, Germany

- Modules: Verification of Digital Systems, Integrated Circuit Design (transistor level), Design of Heterogeneous Systems, Advanced IC Technology, Test and Design of Mixed-Signal Systems.
- **Deutschlandstipendium** 2026/27 – national merit scholarship (industry-sponsored).

B.E. Electronics & Telecommunication Engineering 08/2018 – 04/2022
Visvesvaraya Technological University (BMSIT) Bengaluru, India

- CGPA 8.51 / 10 (German grade ≈ 1.8). **Best Outgoing Student** of the department; state merit scholarship (100 % tuition waiver).
- Modules: CMOS VLSI Design, Computer Architecture, Digital System Design, Microprocessors.

Advanced VLSI Design & Verification – 6-month full-time programme 04/2022 – 09/2022
VLSI training institute Bengaluru, India

- SystemVerilog, UVM testbench architecture, SVA, constrained-random verification, coverage closure, AMBA AXI / AHB / APB, QuestaSim.

TECHNICAL SKILLS

Methodology	UVM (agents, drivers, monitors, scoreboards, sequences), constrained-random verification, functional coverage, verification planning
Assertions	SystemVerilog Assertions (SVA), CDC checks, formal-property concepts, linting
Python verification	cocotb, pyUVM, Spike ISS co-simulation, RISCOF; regression automation and log/coverage analysis
Languages	SystemVerilog, Verilog, VHDL, VHDL-AMS, Python, C / C++, Bash
Simulators & tools	QuestaSim, ModelSim, Cadence Xcelium, Synopsys VCS, Verilator, GTKWave, Design Compiler, Vivado
Protocols & IP	AMBA AXI4 / AHB / APB, PCIe Transaction Layer, Network-on-Chip, RISC-V RV32IM

PUBLICATION

A. Nagaraj et al., “Peripheral Shared Memory Over PCI Express Link,” in *Proc. 7th IEEE International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*, 2025. – Shared-memory throughput, buffer sizing and latency over PCIe.

HONOURS & SELECTED ACTIVITIES

- **NextSEMI 2026 semiconductor hackathon** (SEMICON Europa × TUM Venture Labs, Munich, 11/2026) – one of 25 participants selected across Europe; post-quantum TLS project (see Key Projects).
- **Deutschlandstipendium** 2026/27, TU Chemnitz.
- **Best Outgoing Student** 2022 and state merit scholarship (100 % tuition waiver), VTU / BMSIT.

LANGUAGES

English – fluent (C1) | German – A2 (B1 course in progress) | Kannada – native | Hindi – fluent

FURTHER ACTIVITIES & INTERESTS

Volunteer, IEEE Student Branch TU Chemnitz (technical workshops, semiconductor seminars). Interests: distance running, cricket, hiking in Saxony.

Chemnitz, November 2026